FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – X
PAPER –2
TEST DATE: 21-05-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. B

Sol. $\frac{d^2x}{dt^2} = -\left(\frac{2\rho g x \sin 45° A}{\rho A d}\right)$

$\Rightarrow \frac{d^2x}{dt^2} = -\left(\frac{\sqrt{2}g}{d}\right)x$

Time required = $\frac{T}{2} = \pi\sqrt{\frac{d}{\sqrt{2}g}}$

2. B

Sol. $W_{net} = \Delta K_{cm}$

$F_{floor}\frac{h}{4} + (mg)\left(\frac{-h}{2}\right) = 0$

as the student starts and ends this motion at rest. Therefore, $F_{floor} = 2mg$

3. B

Sol. Energy released = Final binding energy – initial binding energy

$= 110 \times 8.2 + 90 \times 8.4 - 200 \times 7.4 = 178$ MeV

4. A

Sol. From conservation of charge

$-\frac{dq}{dt} = 4\pi r^2 j = 4\pi r^2 \sigma E$

$\Rightarrow -\frac{dq}{dt} = 4\pi r^2 \sigma\left(\frac{q}{4\pi\varepsilon_0\varepsilon_r r^2}\right) = \frac{\sigma q}{\varepsilon_0\varepsilon_r}$

$\Rightarrow q = q_0 e^{-\frac{\sigma}{\varepsilon_r\varepsilon_0}t}$

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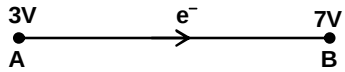
$$\Rightarrow I = \frac{-dq}{dt} = \frac{\sigma q_0}{\varepsilon_r \varepsilon_0} e^{-\frac{\sigma}{\varepsilon_r \varepsilon_0} t}$$

5. A, C

Sol. $-\Delta K = \Delta U$

$K_A - 0 = -e(3-7)$

$K_A = 4eV$

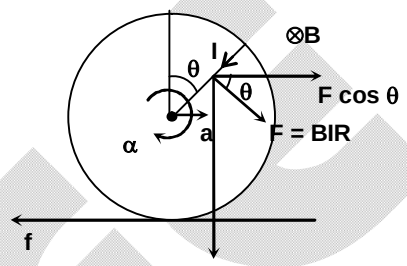


6. A, B, C

Sol. $F\cos\theta - f = ma$ …(i)

$$\left(\frac{FR}{2} + fR\right) = \frac{mR^2}{2}\alpha$$

$$\frac{F}{2} + f = \frac{ma}{2} \quad \text{…(ii)}$$



Adding equation (i) and (ii)

$$F\cos\theta + \frac{F}{2} = \frac{3ma}{2}$$

$$F\left(\frac{2\cos\theta + 1}{2}\right) = \frac{3ma}{2}$$

$$a = (2\cos\theta + 1)\frac{F}{3m}$$

$$\alpha = \frac{F}{3mR}(2\cos\theta + 1) \quad \text{…(iii)}$$

$\omega d\omega = \alpha d\theta$

$$\frac{\omega^2}{2} = \frac{F}{3mR}\int_0^{\pi/2}(2\cos\theta + 1)d\theta$$

$$\omega = \sqrt{\frac{2F}{3mR}\left(\frac{\pi}{2} + 2\right)} = \sqrt{\frac{2BI}{3m}\left(\frac{\pi}{2} + 2\right)}$$

7. A, D

Sol. $V_{rms} = \sqrt{2^2 + \frac{3^2}{2} + \frac{4^2}{2}} = \sqrt{\frac{33}{2}}$ volt

$\Rightarrow V_{av} = 2$ volt

8. B, D

Sol. $\sqrt{D^2 + 4d^2} - D = \frac{\lambda}{2}$

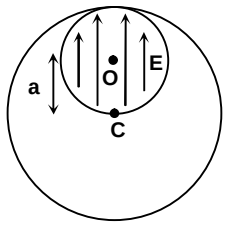
Gives, $\lambda = \frac{4d^2}{D}$

Hence, intensity at P is due to C only, $I = I_0$

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9. A, C

Sol. E in cavity is uniform and it is equal to $E = \frac{\rho a}{3\varepsilon_0}$



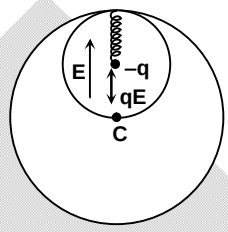
Time period of spring block system is independent of constant force so T $= 2\pi\sqrt{\frac{m}{k}}$

Work energy theorem

$$qEX_{max} - \frac{1}{2}kx_{max}^2 = 0$$

$$X_{max} = \frac{2qE}{k} = \frac{2q}{k}\frac{\rho a}{3\varepsilon_0}$$

$$X_{max} = \frac{2\rho aq}{3\varepsilon_0 k}$$



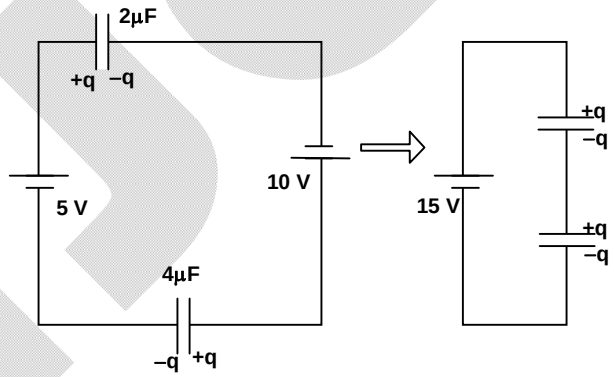
10. A, C, D

Sol. Before the switch is closed, the circuit is shown

Here q = 8/6 × 15 = 20 μC

Total energy stored =

$$\frac{1}{2} \times \frac{20 \times 20}{8/6} = 150\ \mu J$$

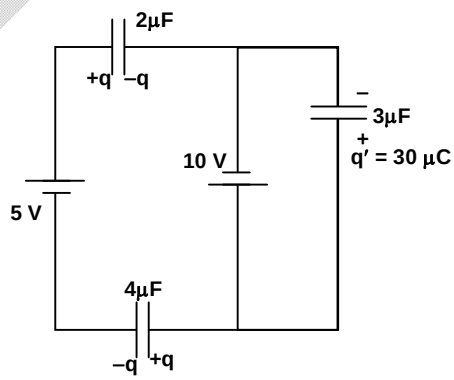


After the switch is closed, the circuit is shown

$$\Delta H = \Delta W_b - \Delta U$$

$$\Delta H = 300 - \frac{1}{2} \times 3 \times 100$$

$$\Delta H = 150\ \mu J$$



Section – B

11. 9

Sol. PV = P′V′

So, that P′ = 0.8P

Since the external pressure is 0.5P, the net outward pressure on the surface of the balloon is 0.3P.

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Thus, the net pressure force on half of the balloon is $0.3\ P\pi R^2$, where R is the final radius of the balloon, given by

$1.25\ V = \frac{4}{3}\pi R^3$

When the balloon bursts,

$\sigma 2\pi Rt = 0.3P\pi R^2$

Here t is the thickness of the balloon material, obtained from

$d = \frac{m}{4\pi R^2 t}$

Combining the equations gives

$\sigma = \frac{9PVd}{16m}$

12. 5

Sol. $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

$\frac{2.4}{2r} - \frac{1}{-d} = \frac{2.4-1}{r}$

$\frac{1}{d} = \frac{1.4-1.2}{r}$

$\frac{1}{d} = \frac{0.2}{r}$

$\Rightarrow d = 5r$

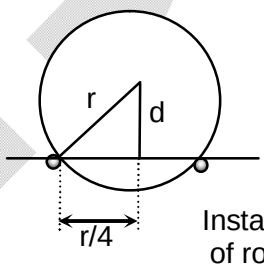
$d = 5\ cm$

13. 5

Sol. $d = \frac{\sqrt{15}}{4}r$

$v = d\omega$

$\omega = \frac{4v}{\sqrt{15}r}$



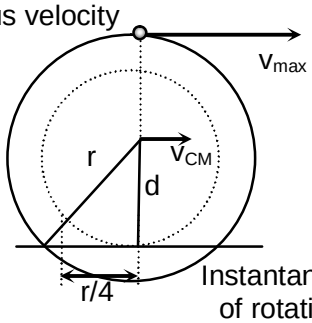
Instantaneous axis of rotation (IAOR)

Front view

Point with maximum instantaneous velocity

$v_{max} = \omega(r + d)$

$v_{max} = \left(\frac{15 + 4\sqrt{15}}{15}\right)v$



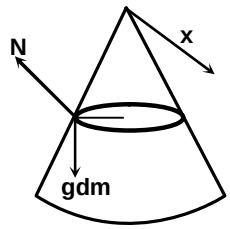
Instantaneous axis of rotation (IAR)

Side view

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14. 4

Sol. Consider the potential energy of the band as a function of R. To be in equilibrium, the band must be at a position of minimum potential energy.



Let the elastic potential energy be $U_{el} = \frac{1}{2}k(2\pi R - 2\pi r)^2$ and the gravitational potential energy measured with respect to the vertex of the cone be $U_g = -mg(R-r)\cot\theta$

The total potential energy is

$U = 2k\pi^2(R-r)^2 - mg(R-r)\cot\theta$

At the minimum position,

$\frac{dU}{dR} = 4\pi^2k(R-r) - mg\cot\theta = 0$

Which has the solution

$R = r + \frac{mg}{4\pi^2k}\cot\theta$

15. 2

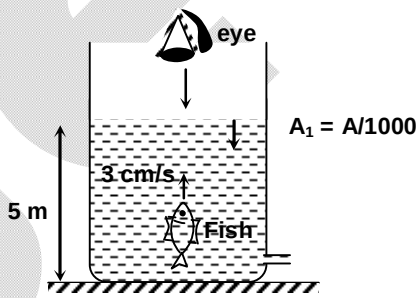
Sol. Velocity of efflux = $v = \sqrt{2g\times 5} = 10$ m/s

Velocity of surface = $\frac{10\times A/1000}{A} = 1$ cm/s

The velocity of fish relative to the observer is

$v = \frac{d}{dt}\left(\frac{x}{\mu} + y\right)$

$= \frac{3}{4}\frac{dx}{dt} + \frac{dy}{dt} = \frac{3}{4}\times(-4) + 1 = -3 + 1 = -2$ cm/s



16. 9

Sol. $\omega_1 = 2\omega_2 = 4\omega_3$, if $\omega_3 = \omega$

then $\omega_1 = 4\omega$, $\omega_2 = 2\omega$

$a_1 = 16\omega^2 r$, $a_2 = 4\omega^2 2r$, $a_3 = \omega^2 3r$

$\Rightarrow$ $a_1 : a_2 : a_3 = 16 : 8 : 3$

17. 3

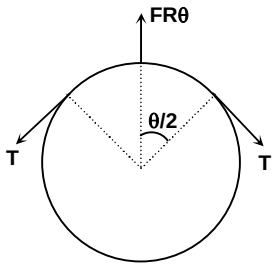
Sol. $\text{Strain} = \frac{T/A}{Y} = \frac{d}{R}$

$\Rightarrow T = \frac{AdY}{R}$

$2T\sin\frac{\theta}{2} = FR\theta$

$T\theta = FR\theta$

$F = \frac{T}{R} = \frac{AdY}{R^2}$



18. 1

Sol. Point C and point D will have same potential.

So, resistance between the points C and D can be removed.

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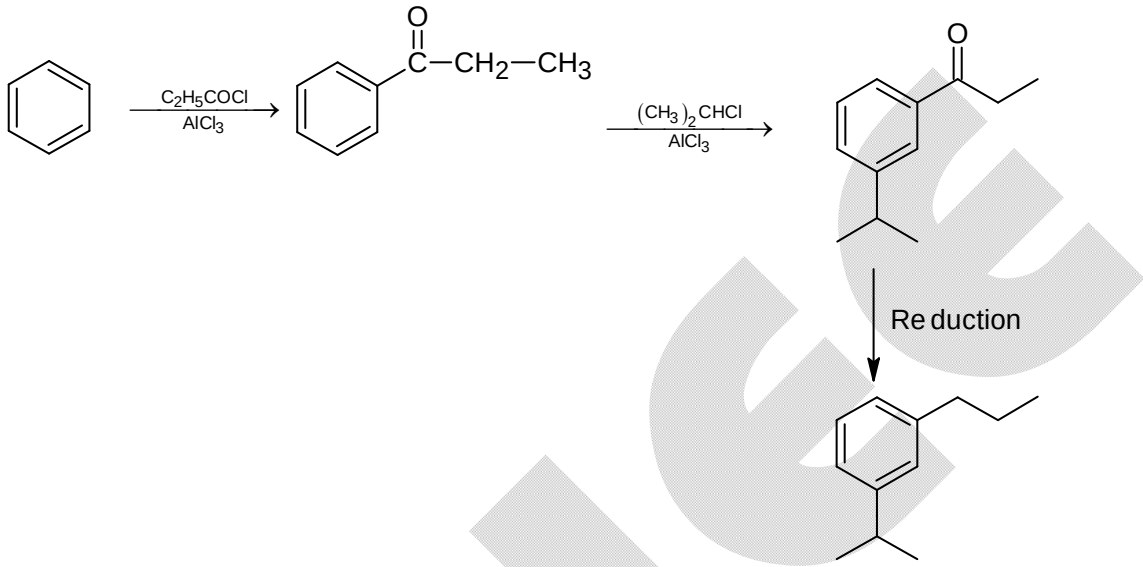
Chemistry

PART – II

Section – A

19. D

Sol.



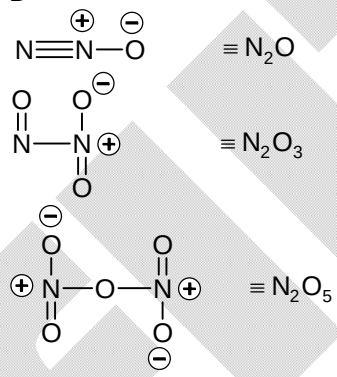
20. B

Sol. $-802.8 = 4 \times 416.2 + 2 \times 493.7 - 2 \times \Delta H_{C=O} - 4 \times 464.4$

$\Delta H_{C=0} = 798.7 \text{ kJ/mol}$

21. D

Sol.



22. D

Sol. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

$x - y$

$C(s) + CO_2(g) \rightleftharpoons 2CO(g)$

$x - y \qquad 2y$

$x - y = 4 \times 10^{-2}$

$\frac{(2y)^2}{(x-y)} = 2 \Rightarrow 2y = \sqrt{2(x-y)}$

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$= \sqrt{2 \times 4 \times 10^{-2}}$

$= 0.28$

$P_{CO} = 2y = 0.28$

23. A, B, C, D

Sol. All these salts react with dil. H_2SO_4.

24. A, B, C, D

Sol. % of N in organic compound $= \dfrac{1.4 \times N \times V}{w}$

$= \dfrac{1.4 \times 10 \times 2 \times 1}{0.5} = 56\%$

Also, milli equivalents of ammonia produced = milli equivalents of H_2SO_4 used

In case of 1 M HCl, the % N $= \dfrac{1.4 \times 10 \times 1}{0.5} = 28\%$

$\%NH_3 = \dfrac{17}{14} \times 56 = 68\%$

25. A, B, C

Sol. On increasing the temperature ionic mobility increases and thus Λ_m increase, so statement (A) is correct.

$\Lambda_m = \dfrac{K \times 1000}{M}$

$\Rightarrow \Lambda_m \propto \dfrac{1}{M}$

Λ_m decreases with increase in concentration. So statement (B) is correct.

As concentration increases, number of ions per unit volume increases. So (C) is correct.

As temperature increases the ionic mobility increases so specific conductance increases. So, statement (D) is incorrect.

26. B, C, D

Sol. $NH_3 + HCl \longrightarrow NH_4Cl$

$NH_3 + CuSO_4 \longrightarrow \underset{\text{Blue}}{\left[Cu(NH_3)_4\right]SO_4}$

$NH_4^+(Salt) + OH^- + K_2[HgI_4] \longrightarrow \underset{\text{Brown colour ppt.}}{I - Hg - O - Hg - NH_2}$

27. B, C, D

Sol. $Cu^{+2} + KCN \longrightarrow K_3\left[Cu(CN)_4\right]$

$Cu^{+2} + NH_3 \longrightarrow \left[Cu(NH_3)_4\right]^{+2}$

$Cu^{+2} + KI \longrightarrow \underset{\text{white ppt.}}{Cu_2I_2} + I_2$

28. A, C

Sol. Cu, Pb and Hg are purified by self reduction.

Ni – Mond process

Ti – van Arkel process

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Section – B

29. 3

Sol. $ZnSO_4 + NaOH(Excess) \longrightarrow Na_2ZnO_2$

$AlCl_3 + NaOH(Excess) \longrightarrow NaAlO_2$

$SnCl_4 + NaOH(Excess) \longrightarrow Na_2SnO_3$

30. 4

Sol. Li, Sr, Ba, Zn

31. 2

Sol. $0.205 = (1+\alpha)0.1\times 1.86$

$1+\alpha = \dfrac{2.05}{1.86} = 1.1$

$\alpha = 0.1$

$\left[H^+\right] = C\alpha = 0.1\times 0.1 = 10^{-2}$

pH = 2

32. 3

Sol. $3B_2H_6 + 18CH_3OH \longrightarrow 6B(OCH_3)_3 + 18H_2$

33. 2

Sol. $mvr = \dfrac{nh}{2\pi}$

$n = \dfrac{4.2178\times 10^{-34}\times 2\times 3.14}{6.625\times 10^{-34}} = 4$

Number of visible line = n – 2

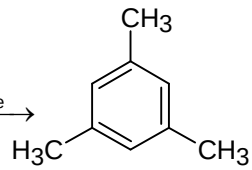
= 4 – 2 = 2

34. 2

Sol. m-methoxy phenol and m-amino phenol.

35. 6

Sol.

$Mg + Coke \longrightarrow Mg_2C_3 \xrightarrow{H_3O^+} C_3H_4 \xrightarrow{Cu-Tube}$ 

[Mesitylene]

36. 4

Sol. HgI_2, HgO, Cu_2O, Pb_3O_4

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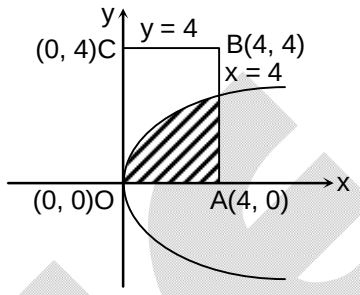
Mathematics

PART – III

Section – A

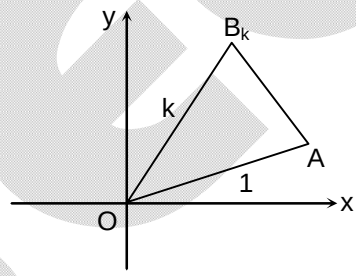
37. B

Sol. Required probability = $\dfrac{\int_0^4 \sqrt{x}\,dx}{16}$



38. A

Sol. $S_k = \dfrac{1}{2}k\sin\left(\dfrac{k\pi}{2n}\right)$



39. D

Sol. $a_{ij} = \lim\limits_{x\to 0}\dfrac{e^{\frac{\ln(1+ix)}{j}}-1}{x} = \dfrac{i}{j}$

40. B

Sol. Differentiate both the sides, keeping x constant

$2xyf'(x^2y) = f(y)\cdot 2x + y\cdot 2x\cdot f'(x^2)$; put x = 1

$yf'(y) = f(y) + yf'(1) = f(y) + y$

Differentiate w.r.t. y

$yf''(y) + f'(y) = f'(y) + 1$

$f''(y) = \dfrac{1}{y}$

41. B, D

Sol. $\dfrac{x^2}{16} - \dfrac{y^2}{4} = 1$; $\left(5, \dfrac{3}{2}\right)$ line on the curve

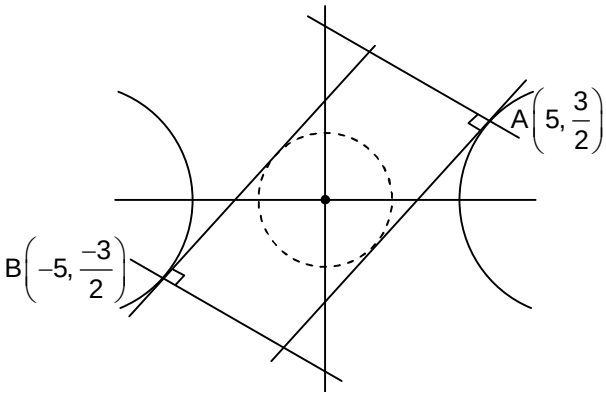
Tangent at A, $\dfrac{5x}{16} - \dfrac{3}{2}\left(\dfrac{y}{4}\right) = 1$

$\Rightarrow 5x - 6y = 16$

Tangent at B, $5x - 6y = -6$

Normal at A, $6x + 5y = \lambda$, $A\left(5, \dfrac{3}{2}\right)$

$30 + \dfrac{15}{2} = 1 \Rightarrow \lambda = \dfrac{75}{2}$; $6x + 5y = \dfrac{75}{2}$



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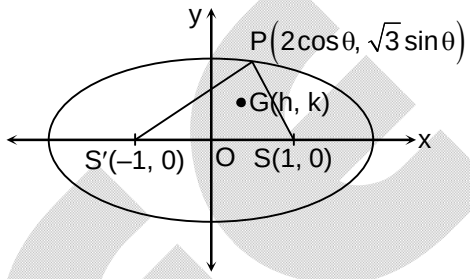
$|||^{ly}$ normal at B is $5x + 5y = -\frac{75}{2}$

Distance between tangents is $2r = \frac{32}{\sqrt{25+36}}$

$\therefore\ r = \frac{16}{\sqrt{61}}$. Radius of circumcircle of rectangle is $R = \sqrt{25+\frac{9}{4}} = \frac{\sqrt{109}}{2}$

42. A, B, C

Sol. $\frac{x^2}{4}+\frac{y^2}{3}=1\ ;\ e^2=1-\frac{3}{4}=\frac{1}{4} \Rightarrow e=\frac{1}{2}$



$\therefore$ 3h = 2 cos $\theta \Rightarrow \cos\theta = \frac{3h}{2}$

and $3k = \sqrt{3}\sin\theta \Rightarrow \sin\theta = \sqrt{3}k$

Now, on squaring and adding, we get

$1=\frac{9x^2}{4}+3y^2$ or $\frac{x^2}{\frac{4}{9}}+\frac{y^2}{\frac{1}{3}}=1$. Now, verify

alternative

43. A, B, D

Sol. (A) For b = 9 solving line and parabola $2x - 9 = x^2 - 4x \Rightarrow x^2 - 6x + 9 = 0 \Rightarrow (x-3)^2 = 0$

$\therefore$ Line is tangent to parabola

(B) For b = 7, line is y = 2x – 7

Homogenising parabola with line $x^2 - 4x\left(\frac{2x-y}{7}\right) - y\left(\frac{2x-y}{7}\right) = 0$

Coefficient of x^2 + coefficient of $y^2 = 0 \Rightarrow 1-\frac{8}{7}+\frac{1}{7}=0$ $\therefore\ \angle AOB$ is 90°

(C) Homogenising the parabola with line, we get $x^2 - 4x\left(\frac{2x-y}{b}\right) - y\left(\frac{2x-y}{b}\right) = 0$

For x-axis to bisect $\angle COD$ coefficient of xy should be zero i.e. $\frac{4}{b}-\frac{2}{b}=0$

Which is not possible for any $b \in R$

(D) For b > 9 solving line and parabola $x^2 - 6x + b = 0$

D = 36 – 4b. If b > 9, then D < 0

$\therefore$ Line passes outside the parabola

44. B, C, D

Sol. Locus of z is $\frac{(x+1)^2}{9}-\frac{y^2}{1}=1$

45. B, C

Sol. $\left(x^{1/2}+\frac{1}{2}x^{-1/4}\right)^n \Rightarrow T_{r+1} = {}^nC_r x^{\frac{n-r}{2}} \cdot \frac{1}{2^r} \cdot x^{\frac{-r}{4}}$

Coefficient of the 1st three terms are ${}^nC_0,\ {}^nC_1 \cdot \frac{1}{2},\ {}^nC_2 \cdot \frac{1}{2^2}$

$\therefore\ {}^nC_0 + {}^nC_2 \cdot \frac{1}{4} = 2 \cdot {}^nC_1 \cdot \frac{1}{2} \Rightarrow 1+\frac{n(n-1)}{8}=n$

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$\therefore \ \frac{n(n-1)}{8} = (n-1) \Rightarrow n = 8$ (as $n \neq 1$)

$\therefore \ T_{r+1} = {}^8C_r \ x^{\frac{8-r}{2}} \cdot \frac{1}{2^r} \cdot x^{-\frac{r}{4}} = {}^8C_r \cdot \frac{1}{2^r} \cdot x^{\left(4-\frac{3r}{4}\right)}$

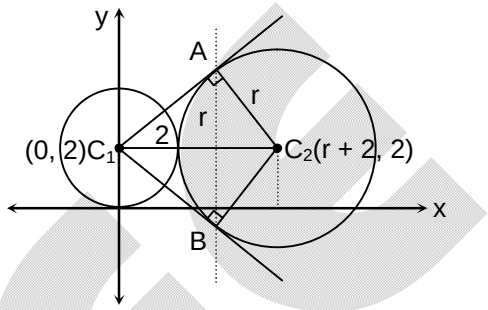
Terms of x with integer power occur when r = 0, 4, 8 $\Rightarrow$ 3 terms

46. A, C

Sol. Area of quad. $C_1AC_2B = 2 \times \frac{1}{2} \times r \cdot 2\sqrt{r+1} = 12$

$\Rightarrow r^2(r+1) = 36$
$\Rightarrow r^3 + r^2 - 36 = 0$
$\Rightarrow r = 3$

Inradius of the circle C_1AC_2 is $= \frac{\Delta}{s} = \frac{\frac{1}{2} \times 3 \times 4}{6} = 1$



Section – B

47. 4

Sol. $\sin 24° = \sin(60° - 36°)$

48. 2

Sol. $V = \frac{1}{3}A_1h_1 = \frac{1}{3}A_2h_2 = \frac{1}{3}A_3h_3 = \frac{1}{3}A_4h_4$

49. 9

Sol. $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1 \Rightarrow \frac{1}{d^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \Rightarrow m = \sqrt{a^2 + b^2 + c^2}$

50. 1

Sol. $Tr = \frac{2r+1}{\left(r(r+1)\right)^2} = \frac{1}{r^2} - \frac{1}{(r+1)^2}$

51. 4

Sol. a, b, c be the sides of ΔABC

52. 8

Sol. Let x be the number of elected person

$\therefore \ {}^{x+1}C_1 + {}^{x+1}C_2 + ... + {}^{x+1}C_x = 254$

53. 2

Sol. Let $(x^2 + 1)e^x = t$

54. 0

Sol. $\left(1 + x + x^2\right)^8 = a_0 + a_1x + ... + a_{16}x^{16}$ ..... (1)

Replace $x \to \frac{1}{x}$

$\left(1 + x + x^2\right)^8 = a_0x^{16} + a_1x^{15} + ... + a_{16}$ ..... (2)

$\therefore$ From equation (1) and (2), we get $a_r = a_{16} - r$

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